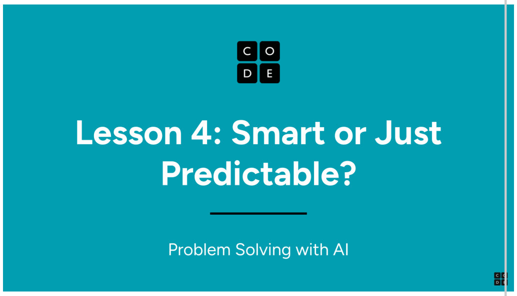
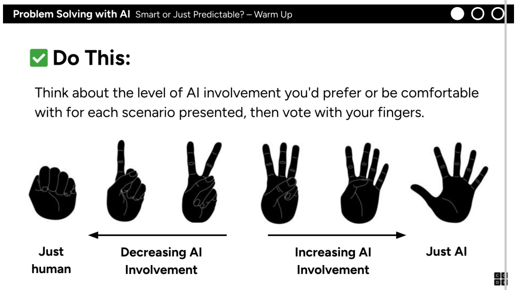
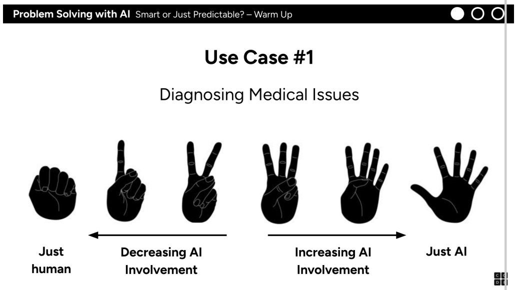
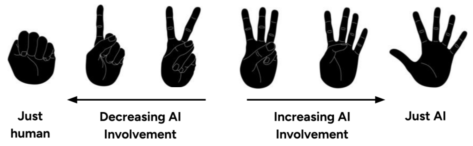
Slide	Notes
	
	Warm Up (10 mins)
	AI Reliability Voting ■ Goal of this activity: Get students thinking about AI's reliability and how much human involvement they would still want in a task that can now be done by AI. ■■ Say: AI is being integrated into many real-world tasks. Some people are excited about this, while others have concerns. ■■ Do This: Inform students that they will be presented with different scenarios on the next few slides. They should think critically about the level of AI involvement they would prefer or be comfortable with for each scenario, then vote with their fingers. Teaching Tip If time allows, you can instead have students line up in your classroom from "Just Human" on one end and "Just AI" on the other and give them a minute to discuss their reasons for where they lined up along the spectrum with those around them.
	■■ Do This: Present the first real-world AI use case: diagnosing medical issues. Ask students to vote on the level of AI involvement they would prefer for this use case. After students vote, ask a couple students from each option (0-1 finger, 2-3 fingers, 4-5 fingers) to explain their reasoning. Students should recognize that AI has strengths but also clear limitations that require human oversight. The goal is to try to get students to begin considering what makes AI reliable vs. risky in different applications. Teaching Tip If students struggle to think critically about the risks vs. benefits, prompt them with follow-up questions, such as "What's the worst-case scenario if AI makes a mistake in this task?" or "How could AI and humans work together in this situation?"

Use Case #2

Grading Essays



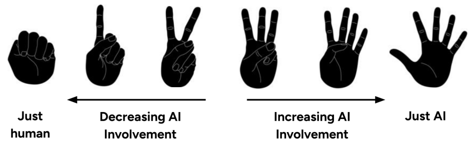
■■ Do This: Present the second real-world AI use case: grading essays. Ask students to vote on the level of AI involvement they would prefer for this use case. After students vote, ask a couple students from each option to explain their reasoning.

Teaching Tip

If students say AI isn't useful for grading, ask: "Could AI help teachers save time?" or "If AI misses deeper meaning in writing, how could we still use it effectively?" Bring up tools like Grammarly—AI helps with mechanics, but teachers provide feedback on ideas.

Use Case #3

Driving a School Bus



■■ Do This: Present the third real-world AI use case: driving a school bus. Ask students to vote on the level of AI involvement they would prefer for this use case. After students vote, ask a couple students from each option to explain their reasoning.

Teaching Tip

If students say AI should fully replace bus drivers, ask: "What happens if a child runs into the street unexpectedly?" or "Can AI handle a traffic situation as well as a human driver?" Connect this to real-world examples like Tesla's autopilot mode—it assists but doesn't replace drivers.

Lesson Objectives

By the end of this lesson, you will be able to . . .

- Compare AI and human performance to identify AI's strengths and weaknesses.
- Evaluate AI's reliability and when human oversight is needed.
- Justify when AI should or shouldn't be used in real-world cases.

■■ Do This: Introduce the lesson objectives.

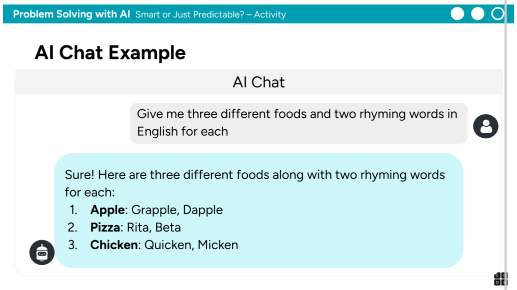
Question of the Day

Where does AI excel vs struggle?

■■ Do This: Introduce the Question of the Day.

Activity

Activity (30 mins)

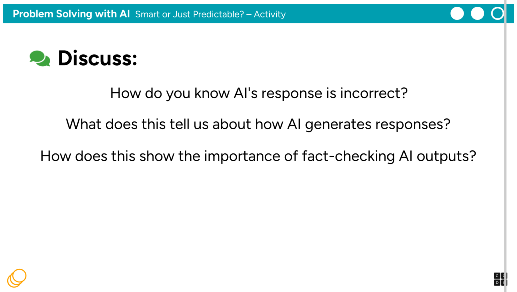


AI Hallucination Example and Discussion (10 mins)

■ Goal of this activity: Introduce students to AI-generated hallucinations and why AI gets things wrong.

■■ Do This: Ask students to analyze where AI went wrong. Prompt them to identify errors in AI's response and discuss why they happened. Encourage students to look for patterns in the errors. Give students a minute to share with a partner before sharing as a class.

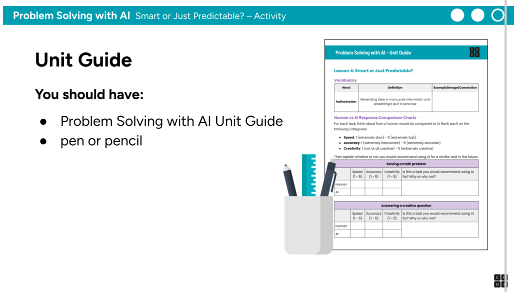
■ Circulate: Listen for students pointing out how AI generates words based on patterns rather than understanding. If students struggle to identify why "Micken" or "Beta" is incorrect, ask: "What pattern do you see in AI's mistakes? How does AI choose words?"



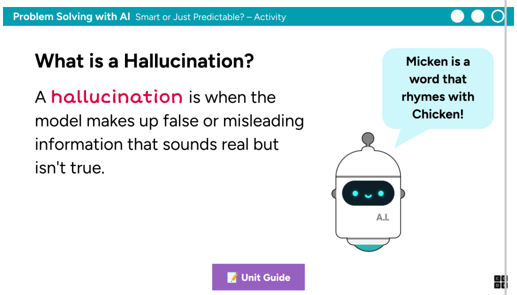
■ Discuss: Click through the animated slide to display the prompts.
How do you know AI's response is incorrect?
What does this tell us about how AI generates responses?
How does this show the importance of fact-checking AI outputs?

■ Discussion Goal: Students should understand that AI predicts based on probability, not actual knowledge. The goal is to get students to begin considering why AI makes errors and when it should be fact-checked. Example student responses may include:
"AI doesn't actually know meanings; it just predicts words that commonly follow each other"
"It made up words because it's trying to complete a pattern rather than understanding the meaning"
"That means AI can make errors in more serious situations too, like if it made up medical information."

Teaching Tip
If students are struggling to articulate why AI makes mistakes, prompt them with: "Does AI understand language like a human, or is it just following patterns?" or "What would happen if we trusted AI without checking its facts?"



■■ Do This: Have students take out their Problem Solving with AI Unit Guide.



■■ Say: AI doesn't "think" like a human – it predicts based on probability and past data. AI can make up words, facts, or quotes because it's trying to complete a pattern, not verify truth. This occurs when the most probable response isn't accurate or factual.

■■ Do This: Define hallucination.

Teaching Tip
Help students connect this definition to real-world AI failures. Explain that AI doesn't know "Micken" isn't a real word—it just predicts what's most likely to come next based on training data. This is why AI sometimes generates confident but completely false responses.

Problem Solving with AI Smart or Just Predictable? – Activity

Fact-Checking AI – Why It's Important

- AI doesn't verify information – it predicts what is most likely to sound correct.
- Humans must fact-check AI-generated responses before trusting them.
- AI-generated mistakes (hallucinations) can seem correct if not closely examined.
- AI's predictions can lead to misinformation if not fact-checked.

■■ Do This: Click through the animated slide to explain that AI doesn't verify information before providing a response - it just outputs what it predicts is most likely to sound correct. This is why humans should fact-check any AI-generated responses before trusting them, even when the hallucination seems plausible. Not fact-checking can lead to misinformation.

Teaching Tip
You do not need to go in-depth into misinformation and fact-checking as this will be covered in an upcoming lesson. Use this as an opportunity to get students to start thinking about the importance of fact-checking.

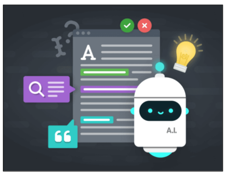
Problem Solving with AI Smart or Just Predictable? – Activity

AI vs Human Tasks

Lesson 4 1

Do This:

1. Go to Lesson 4, Level 1.
2. Test AI's ability with different tasks on Levels 1, 2, and 3.
3. Complete the **Human vs AI Response Comparison** section in your Unit Guide.



Unit Guide

AI vs Human Tasks (20 mins)

■ Goal of this activity: Allow students to experiment with AI capabilities and determine where AI performs well vs. where it struggles. Encourage students to compare AI-generated vs. human-generated responses to develop the Critical Consumer mindset.

■■ Say: We just saw an example of AI making up words. Now, we're going to test AI with different types of tasks to see what it does well and where it struggles.

■■ Do This: Group students into pairs for the purpose of having a thought partner to share experiences and ranking with for the next activity.

■ Transition: Direct students to Lesson 4, Level 1 to complete Levels 1 through 3 to test AI's performance on different tasks. Students should complete the "Human vs AI Response Comparison" section of their Unit Guide as they interact with each task. Have students also complete the quick question on Level 4.
Level #1: Students test AI's ability to solve a math problem. They rate how well a human would do and how the AI did.
Level #2: Students test AI's ability to answer a creative or tricky question. They rate how well a human would do and how the AI did.
Level #3: Students test AI's ability to provide historical or technical information. They rate how well a human would do and how the AI did.
Level #4: Students answer a quick questions related to careers in computing science.

■ Circulate: Ensure students are sharing and comparing results and rankings for each task with their thought partner. Observe how students are ranking AI vs human performance in the AI vs Human Response Comparison. If students are unsure, ask: "What criteria are you using to decide whether AI or a human is better at this task?"

Teaching Tip
Each level has multiple options for students to choose from as tasks to provide to the AI. If time allows, encourage students to test multiple types of tasks to see if their rankings remain consistent

Problem Solving with AI Smart or Just Predictable? – Activity

Discuss:

What tasks was AI best at? Why?

Where did it struggle? Why?

■ Discuss: Click through the animated slide to display the prompts.
What tasks was AI best at? Why?
Where did it struggle? Why?

■ Discussion Goal: Students should recognize that AI excels at structured, data-driven tasks but struggles with creativity and reasoning. Aim to get students to connect AI's limitations to its training data and probability-based responses.

Teaching Tip
If students struggle to identify patterns in AI's performance, ask: "What did AI do well across multiple tasks?" or "Did AI struggle with logic-based tasks or tasks that required creativity?"

Problem Solving with AI Smart or Just Predictable? – Activity

AI vs. Humans – Strengths and Weaknesses

- AI is **great at structured, data-driven tasks** (e.g., solving math).
- AI **struggles with reasoning, creativity, and common sense**.
- AI **may provide misinformation or fake quotes** that sound real.

■■ Do This: Explain that AI is really good at things like math, but it doesn't have human intuition or logic. Due to hallucinations, it needs to be fact-checked.

Teaching Tip
If students assume AI is better just because it's faster, ask: "When does accuracy matter more than speed? Can AI make fast but incorrect decisions?"

Problem Solving with AI Smart or Just Predictable? – Activity

Discuss:

How did AI's responses compare to human-generated responses?

Would you trust AI's responses without fact-checking?

Why or why not?

■ Discuss: Click through the animated slide to display the prompts.
How did AI's responses compare to human-generated responses?
Would you trust AI's responses without fact-checking?
Why or why not?

■ Discussion Goal: Students develop the Critical Consumer mindset by questioning where AI-generated content needs human oversight.

Teaching Tip
If students appear to struggle to make the connection between hallucinations and the tasks they tested, ask the follow up questions: "How does what we just tested compare to the hallucination example? Did AI make similar mistakes?"

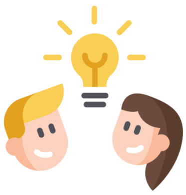
Problem Solving with AI Smart or Just Predictable? – Activity

Check Your Understanding

Lesson 4 ○○○ 5 ○

Do This:

1. Go to Lesson 4, Level 5.
2. Respond to the questions.



■■ Say: Now that we've ranked AI's performance, let's think about how this applies to real-world careers. Do we trust AI to handle critical tasks in medicine, law, travel, or writing?

■ Transition: Direct students to Lesson 4, Level 5 to complete the formative assessment.

■■ Do This: Display students' anonymous responses and ask students "What do you notice?" and "What do you wonder?" Have students discuss with a partner before asking for a couple volunteers to share.

Teaching Tip
If students only focus on AI's benefits, ask: "What are the risks if AI makes a mistake in this career?" or "Would an AI error have minor or serious consequences in this field? Why?"

Assessment Opportunity
This level can be used as a formative assessment to check student understanding of the lesson objectives. Here are things to look for and actions you can take:
If students assume AI can fully replace humans, revisit the AI vs. Humans – Strengths and Weaknesses slide to reinforce AI's limitations.
If students struggle to justify their reasoning, prompt them to connect their responses back to their AI vs. Human Response Comparison rankings.
If students demonstrate strong understanding, move on to the wrap-up discussion.

Wrap Up

Wrap Up (5 mins)

Today, you learned about . . .

- Comparing AI vs human performance to identify AI's strengths and weaknesses.
- AI's reliability and when human oversight is needed.
- When AI should or shouldn't be used.

■■ Do This: Review the lesson objectives.

Question of the Day

Where does AI excel vs struggle?

📝 Unit Guide

Question of the Day Reflection

■ Goal of this activity: Ensure all students reflect on AI's strengths and weaknesses as a collaborator by sharing their insights in small groups. Reinforce the Critical Consumer mindset by encouraging students to verbalize when AI is useful vs. when human reasoning is needed.

■■ Do This: Direct students to answer the Question of the Day in their Unit Guide.

Assessment Opportunity
If students struggle to differentiate AI's strengths from its weaknesses, revisit the AI vs. Humans – Strengths & Weaknesses slide and discuss specific examples.
If students make strong, specific comparisons, move on to the Pass the Mic activity.

✅ Do This:

1. Take turns answering the Question of the Day in one or two sentences.
2. Then "pass the mic" so that everyone in your group has a chance to share.

■■ Do This: Have students form small groups of three or four.

■■ Do This: Explain that now that they have reflected in their Unit Guide, they will share key takeaways with their group. Explain that the "Pass the Mic" discussion is designed so that everyone gets a chance to share. Each person will share their answer to the Question of the Day in one or two sentences and then "pass the mic" so that everyone has a turn.

■ Circulate: As students share in their groups:
Listen for common themes and misconceptions such as:
Are students clearly identifying AI's strengths and weaknesses?
Are students misunderstanding AI's capabilities (e.g., assuming AI "thinks" like a human)?
Prompt deeper thinking with guiding questions such as:
Did AI ever surprise you with what it could or couldn't do?
Would you trust AI more or less after today's lesson?
Did anyone in your group have a different perspective that made you think differently?

Teaching Tip
If time allows, facilitate a full-class discussion of the QotD. Consider asking students to raise their hands if they heard a new perspective in their group that made them think differently.
If students only state a general opinion (e.g., "AI is good at math"), follow up with: "What makes AI good at that task? Is it better than a human?" or "Can you give an example of when AI would fail at that same task?"
If some students are not engaging in discussion, ask them directly: "What did you write down in your Unit Guide? Can you share one thing that stood out to you?" or "Did you hear something from your group that you agree or disagree with?"

Key Vocabulary

- **hallucination:** when the model makes up false or misleading information that sounds real but isn't true

■■ Do This: Review the Key Vocabulary.